

BREAK INTO AEROSPACE

Rolls-Royce

Engineering Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Rolls-Royce Holdings is one of the world's leading aero-engine manufacturers, producing and supporting civil and military gas turbines across a global fleet of over 13,000 engines. Its TotalCare business model — which ties revenue directly to engine availability across a 30-year service life — makes Rolls-Royce uniquely accountable for the whole-life engineering performance of everything it designs. In 2025–2026 the company is in a sustained period of commercial recovery and growth, investing heavily in UltraFan, SAF compatibility, and next-generation engine capability. This guide consolidates the preparation tasks, signal areas, and strategic context needed to compete effectively across all relevant engineering roles.

PART I · WHAT ROLLS-ROYCE ACTUALLY LOOKS FOR

By Seniority Level

- **Intern / Industrial Placement (Year in Industry)** — Rolls-Royce's early talent pipeline is selective and high-calibre by design. Assessment centre attendees are typically extremely high-calibre — meeting people with multiple prior internships is common — and the calibre of the group is noticeable from the moment candidates arrive. At intern level, Rolls-Royce is not expecting domain expertise — it is screening for strong academic grounding in a relevant engineering discipline, demonstrable curiosity about how things work physically (thermodynamics, materials, structures, systems), and an ability to function in a structured, process-governed manufacturing and development environment. The ability to work collaboratively and communicate across levels is assessed directly through the group exercise format. *Careers in Aerospace*

One context that matters for 2025–2026 specifically: Rolls-Royce is in a sustained period of recovery and growth, posting a 74% rise in underlying pre-tax profit in the first half of 2025, with UltraFan ground testing in Derby proving successful. The company is investing heavily in engineering capability and hiring actively. Interns entering now are entering an organisation in commercial momentum — but one that remains deeply engineering-conservative in its culture. *Rollsroycefirstnetwork*

- **Mid-Level (Graduate through ~5 years)** — At mid-level, Rolls-Royce is hiring engineers who can sit within an established engineering discipline — combustion, fan aerodynamics, structures, controls, systems, materials, manufacturing — and contribute meaningfully to a programme that has decades of institutional knowledge behind it. The assessment centre at this level includes a competency interview with a self-presentation, STAR-based behavioural questions, a technical interview focused on problem-solving and application of academic knowledge to practical projects, and a group exercise assessing team dynamics. The technical interview at this level specifically probes whether you can reason from fundamentals — a common format involves being handed a physical engine component and asked to explain what it does, where it sits in the engine, how it was likely manufactured, and what it might fail from. *Rolls-Royce*

- **Senior (5+ years, specialist or technical leadership track)** — Rolls-Royce engineers at senior level sit on one of two tracks: the technical specialist pathway (deep domain expertise, contributing to certifica-

tion evidence, writing engine standards, acting as subject matter authorities) or the technical leadership pathway (programme ownership, cross-discipline integration, supplier management). At senior level the specific filter is whether your technical judgement is trustworthy enough to defend an engineering position to EASA or the CAA, and whether you can build and maintain the institutional knowledge that makes the Trent family's decades-long support commitment viable. Senior engineers at Rolls-Royce are not expected to be generalists — they are expected to be the deepest expert on something specific and to have demonstrated it in a regulated context. TotalCare contracts, which cover over 90% of the Trent fleet on a power-by-the-hour model, mean Rolls-Royce commits to engine performance and availability over the entire service life — senior engineers carry the technical accountability that makes that commitment credible. *In Practise*

Hands-On Signal Areas

Domain	Core Competencies Expected
Gas Turbine Domain Knowledge	Rolls-Royce's aerospace business is almost exclusively gas turbines. Technical interviews have specifically included questions about jet engine components, the thermodynamic cycle, and materials — including being handed a turbine blade and asked where in the engine it sits and how it was manufactured. Engineers who have not engaged seriously with the Brayton cycle, turbomachinery, or turbine blade materials are at a structural disadvantage. <i>Graduates First</i>
EASA Part 21 / CS-E Familiarity	Rolls-Royce designs, tests, builds, and certifies engines to EASA and FAA standards. Engineers who have touched a certification programme, even peripherally, are distinctly more attractive than those who have not. <i>Avitrader</i>
Materials & Manufacturing Process Awareness	Rolls-Royce uses advanced single-crystal alloys, ceramic matrix composites, titanium aluminides, and complex additive manufacturing processes. Engineers who understand why specific materials are used in specific engine locations — and what their failure modes are — demonstrate the depth of thinking that Rolls-Royce development engineering requires.
TotalCare & In-Service Data Experience	Because Rolls-Royce services engines on long-term contracts, in-service data analysis, Engine Health Monitoring (EHM), and whole-life cost thinking are everyday engineering disciplines. Engineers who have worked with field data, not just design calculations, are valued.
UltraFan & Sustainability Fluency	UltraFan, which delivers a 10% efficiency improvement over the Trent XWB and is 100% SAF-compatible, is the most significant new engine design in decades. At any level, being able to discuss the geared architecture, the advanced core, and the thermodynamic efficiency gain is a signal that you are paying attention to the engineering that matters to the company's future. <i>Aerospace Manufacturing</i>
Whole-Engine Integration Experience	Rolls-Royce values engineers who have thought about how their component interacts with the rest of the engine — thermal loads passing between modules, vibration modes coupling between fan and core, control system interactions with engine transient behaviour.

How Rolls-Royce's Interview Format Differs from a Standard Technical Interview

- **The assessment centre is the dominant format and it is comprehensive.** The Rolls-Royce assessment centre includes: a group exercise assessing teamwork, a competency interview beginning with a 10-minute self-presentation followed by STAR-based behavioural questions, a technical interview with problem-solving and domain reasoning, and a case study or written exercise depending on the stream. The full day runs approximately seven hours. All three strands (group, competency, technical) carry real weight — candidates can be strong in one and still not receive an offer. *Rolls-Royce*
- **The technical interview is hands-on and tactile in a way that is specific to Rolls-Royce.** Interviewers have given candidates a physical gas turbine component — a turbine blade, a fan blade, a combustion liner — and asked them to explain its function, its location in the engine cycle, and how it was likely manufactured. This is not an exercise you can prepare for purely through textbooks. It requires actual familiarity with gas turbine components, their visual characteristics, and the engineering logic behind their design. *Glassdoor*
- **The self-presentation is scored and must address how you add value to Rolls-Royce specifically.** The competency interview opens with a 10-minute self-presentation covering who you are, your skills, and how you would add value to your role within Rolls-Royce. This is not a career biography — it is an argument. Candidates who deliver a chronological tour of their CV instead of a structured case for why they are the right engineer for this specific team score poorly. *Rolls-Royce*
- **The group exercise is observed for collaboration quality, not just output.** Assessors are watching listening behaviour, yielding, challenge quality, and constructive contribution — not just whether the group reaches a correct answer. Dominating the group is penalised as clearly as disengaging from it. *Rolls-Royce*
- **Online psychometric tests are a hard pre-screen gate.** Before the assessment centre, candidates complete numerical, verbal, and logical reasoning tests, and situational judgement tests. These are not formalities — they thin the applicant pool significantly before any human reviews a CV. *Graduates First*
- **For experienced professional hires, the format is different.** Experienced hires typically go through a recruiter screen, a technical interview with the hiring manager (structured around a project walk-through), and a final panel interview. The assessment centre format is generally reserved for graduate and intern selection. However, the same technical depth expectations apply — and the physical component reasoning that appears at assessment centre level reappears in a more sophisticated form at experienced hire level.

Honest caveat on format variation: Rolls-Royce's UK aerospace engineering is concentrated in Derby (civil aerospace engine development and manufacturing) and Bristol (military aero-engines — Typhoon, F-35 engine components). Rolls-Royce's nuclear submarine propulsion work is in Derby and Bristol but is a separate business unit. The technical depth and domain focus differ significantly between civil and defence streams. If you know which division you are targeting, focus your technical prep accordingly.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Placement Level

1. Build a STAR Answer Bank Against Rolls-Royce's Specific Behavioural Competencies

3–4 h

Skill: Competency interview preparation for an engineering-culture behavioural format

Objective: The competency interview uses STAR-format questions following the self-presentation. Rolls-Royce's published competencies include: collaborative working, innovation, results focus, communication, and professional integrity. An intern whose STAR stories demonstrate a preference for working alone, or who conflates "results" with individual achievement rather than team contribution to a defined engineering outcome, will not land well. Rolls-Royce is specifically selecting for engineers who will function well in a team-based, process-governed, long-programme environment.

Done Signal: You have five written STAR stories (one per competency: collaboration, innovation, results, communication, integrity), each drawn from real experience, each ending with a specific and measurable outcome. Each story has been spoken aloud to another person who confirmed the competency was evident in the content, not just asserted.

Source: Rolls-Royce

2. Learn the Brayton Cycle and Gas Turbine Engine Architecture to the Point of Explanation Without Notes

4–5 h

Skill: Gas turbine thermodynamics and physical engine literacy

Objective: Technical interview questions have specifically covered jet engine components, thermodynamic cycle reasoning, and materials — including being handed a physical component and asked to explain it. An engineering candidate who cannot explain the Brayton cycle, why each engine module exists, and what happens thermodynamically through each stage cannot engage with the technical interview credibly. This is not advanced turbomachinery — it is the foundational vocabulary of the environment you are joining.

Done Signal: You can draw a cross-section of a two-shaft high-bypass turbofan from memory, label every major module (fan, IP compressor, HP compressor, combustor, HPT, LPT, nozzle), explain the thermodynamic process at each stage with T-s diagram references, and explain why bypass ratio is relevant to fuel efficiency. You can do this in under 5 minutes without notes.

Source: Graduates First

3. Prepare a 10-Minute Self-Presentation Structured as an Argument, Not a Biography

3–4 h

Skill: Scored self-presentation making a specific case for your value

Objective: The self-presentation covers who you are, your skills, and how you would add value to your role within Rolls-Royce. The framing question is "add value" — not "here is my history." Candidates who deliver a chronological CV tour give the assessor no reason to prefer them over another technically similar candidate. The presentation must be structured as a case: here is the specific engineering challenge the team faces, here is what I know and have done that is relevant to it, here is what I will contribute.

Done Signal: You have a 10-minute presentation that opens with the role's engineering context (not your biography), builds through your three strongest evidence points, and closes with a specific statement about what you would contribute in the first six months. You have delivered it twice to different audiences and remained within the time limit both times.

Source: Rolls-Royce

4. Practise Identifying a Physical Component by Function, Cycle Position, and Likely Manufacturing Process 4–5 h

Skill: Physical gas turbine component reasoning

Objective: Candidates at Rolls-Royce assessment centres have been handed a turbine blade, fan blade, or combustion liner and asked to reason about it out loud — where it is in the engine, what it does, what loads it sees, and how it was likely made. This exercise cannot be prepared for through theory alone. You need to have looked at, touched if possible, and reasoned about real engine components.

Done Signal: You have studied images of at least six distinct engine components (fan blade, compressor blade, turbine blade, combustion liner, nozzle guide vane, disc, seal, bearing housing), can identify each by visual appearance alone, and can explain for each: its position in the engine cycle, the primary loads it experiences, the dominant failure mechanism, and the most likely manufacturing process (casting, forging, machining, additive, composite lay-up).

Source: Glassdoor

5. Complete Numerical, Logical, and SJT Practice Assessments Under Timed Conditions 4–6 h

Skill: Pre-screen psychometric test performance

Objective: Online assessments including data analysis, SJTs, and numerical tests precede the assessment centre and thin the candidate pool before any human review of a CV. Candidates who underperform on psychometric tests do not reach the assessment centre regardless of their technical strength.

Done Signal: You have completed at least three full-length SHL-style numerical reasoning tests and two SJT sets under timed conditions with scores at or above the 75th percentile, and have identified and practised your weakest question types specifically.

Spread over multiple days. Source: Graduates First

6. Practise a Group Exercise with Live Participants — and Ask for Feedback on Listening and Yielding 3–4 h

Skill: Observed collaborative problem-solving

Objective: The group exercise specifically assesses how you work as a team, with assessors watching how candidates listen, contribute, and collaborate — not just whether a solution is reached. The behaviours that assessors penalise — dominating, dismissing others' ideas, disengaging — are rarely visible to the candidate in the moment. You cannot identify these habits without live feedback from observers.

Done Signal: You have run at least two live group exercises (use engineering case studies or business problem prompts) with 4–6 people, and at least one observer has given specific, named feedback on your listening, yielding, challenge quality, and final contribution to the group's output. You have corrected at least one identified behaviour in a second practice.

Source: Rolls-Royce

Mid-Level (Graduate through ~5 years)

All intern tasks still apply. The self-presentation (task 3) must be built around professional project experience, not academic projects. The component reasoning (task 4) must extend to explaining failure modes from a materials engineering perspective, not just function.

7. Study the EASA CS-E Certification Specification and Understand What a “Type Certificate” Means for an Engine 5–6 h

Skill: EASA engine certification framework literacy

Objective: Rolls-Royce designs, tests, builds, and certifies engines to EASA and FAA standards — the Pearl 700 certification process involved close collaboration with EASA across multiple test locations globally. Mid-level engineers working on engine development are part of, or adjacent to, a certification programme. Not understanding what a Type Certificate is, what CS-E requires for structural integrity or bird ingestion, or what the difference between a Design Organisation Approval (DOA) and a Production Organisation Approval (POA) means, signals a gap that is directly visible in daily work.

Done Signal: You can explain what an EASA Type Certificate covers for a civil turbofan engine, name five certification specifications from CS-E that drive specific test requirements (bird ingestion, fan blade-off, icing, surge margin, endurance), and explain the difference between a DOA and a POA and why both are necessary for Rolls-Royce to design and produce engines.

Source: Avitrader

8. Prepare a Story Demonstrating You Have Contributed to or Engaged with a Formal Engineering Review or Gated Development Process 2 h

Skill: Gate-based development process participation

Objective: Rolls-Royce develops engines through a structured technology readiness and design maturity gating process — each gate requires specific evidence of design maturity before moving forward. Internal processes including engineering gateways and technology readiness levels govern R&D investment allocation. Mid-level candidates who have only worked in informal or Agile-only development environments without formal review gates will be visibly unfamiliar with how development decisions are made and justified at Rolls-Royce.

Done Signal: You have a story involving a formal review gate (design review, milestone review, test readiness review, or equivalent) where the review required specific evidence of maturity, the evidence package was prepared and presented formally, and the outcome either cleared the gate or resulted in a documented action plan. The story names the specific maturity criteria and the evidence provided.

Source: In Practise

9. Develop Working Knowledge of at Least One Advanced Engine Material and Its Engineering Trade-Offs 4–5 h

Skill: Engine materials engineering depth — one specific domain

Objective: Materials engineering is central to Rolls-Royce’s competitive position. Advanced single-crystal alloys, ceramic matrix composites, titanium aluminides, and additive manufacturing processes are used across engine modules — each chosen because of specific thermomechanical properties balanced against cost, weight, and repairability. Mid-level engineers who can speak only about generic steel or aluminium, without engaging with the specific material challenges of the high-pressure hot section or the fan system, are demonstrating a gap in Rolls-Royce-relevant depth.

Done Signal: For one advanced engine material (e.g., single-crystal nickel superalloy, ceramic matrix composite, titanium aluminide, carbon fibre fan blade), you can explain: the thermomechanical properties that make it suitable for its engine application, the primary failure modes, the manufacturing process and its challenges, and the key engineering trade-off that governs when this material is chosen over an alternative. You can answer one hostile follow-up question without hesitation.

Source: Rolls-Royce

10. Prepare a Story of Improving Reliability or Reducing Cost on a Product or Process You Have Owned 2 h

Skill: Reliability improvement and cost awareness — connected to TotalCare context

Objective: TotalCare contracts cover over 90% of the Trent fleet on a fixed fee-per-engine-flying-hour model, meaning every hour of unscheduled maintenance, every early removal, and every shop visit that was not predicted costs Rolls-Royce directly. This creates a deeply embedded culture of reliability engineering and life-cost thinking. Mid-level engineers who have never thought about whole-life cost, unscheduled removal rate, or reliability growth are missing a framing that Rolls-Royce applies to every engineering decision.

Done Signal: You have a story where you improved the reliability, durability, or cost of a product or process — you can name the metric that changed (MTBF, scrap rate, inspection interval, shop visit rate, life limit), the engineering change you made, the data that confirmed the improvement, and the financial or operational consequence of having made it.

Source: In Practise

11. Understand the TotalCare Business Model and Its Engineering Implications at a Working Level 2–3 h

Skill: Service-business engineering context — specific to Rolls-Royce's commercial structure

Objective: Rolls-Royce generates roughly 50–60% of its group revenue from services, driven principally by TotalCare in civil aerospace — a fixed fee-per-flying-hour model that aligns incentives between Rolls-Royce and airlines and generates predictable, recurring revenue tied to utilisation. The engineering consequence of this model is profound: every design decision affects not just the development cost of the engine but its whole-life cost over 30+ years of service. Engineers who do not understand that their work is ultimately governed by this economic model cannot make fully informed engineering trade-offs.

Done Signal: You can explain what TotalCare is, why it aligns Rolls-Royce's incentives with airline operational reliability (not just engine delivery), how a reduction in shop visit interval affects Rolls-Royce's cost base, and how this drives engineering priorities differently from a pure OEM-sales model. You can connect this to at least one engineering trade-off in your own domain.

Source: In Practise

12. Prepare a Story About Working with a Supplier or External Partner — Including a Case Where They Did Not Meet a Specification 1–2 h

Skill: Supplier management and non-conformance handling in a manufacturing context

Objective: Rolls-Royce is an engine integrator that relies on a complex global supply chain for components, raw materials, and subsystems. Mid-level engineers routinely interact with suppliers — reviewing their design proposals, witnessing their tests, dispositioning their non-conformances. Candidates who have only done in-house development without supplier interface are missing a real dimension of Rolls-Royce engineering work.

Done Signal: You have a story where you reviewed a supplier's deliverable (component, analysis, test report, drawing), identified a gap against a specification or requirement, formally documented the non-conformance or observation, and tracked the resolution to closure. The story names the specification that was at issue and the disposition method.

13. Prepare a Specific, Technically-Grounded Answer to “Why Rolls-Royce?” That Addresses Both the Company’s Current Strategic Context and Your Own Technical Trajectory 2–3 h

Skill: Company-specific motivation that survives a technical follow-up

Objective: Rolls-Royce specifically assesses motivation for the company during the self-presentation and competency interview. Competency questions include questions about Rolls-Royce directly, and the self-presentation specifically asks how you would add value. A “Why Rolls-Royce?” answer that says “I’ve always admired the brand” or “I want to work on cutting-edge technology” is answered correctly only when it names a specific technical challenge (UltraFan’s geared architecture, SAF combustion optimisation, TBC thermal management on HPT blades) and connects it to what the candidate has actually studied or worked on.

Done Signal: You have a 90-second answer that names a specific technical programme or challenge at Rolls-Royce, explains the engineering significance of that challenge in your own words, connects it specifically to your technical background, and can answer one follow-up question about the technical challenge without going vague.

Source: Rolls-Royce

Senior Level (5+ years, specialist or technical leadership)

All mid-level tasks still apply. The EASA CS-E knowledge (task 7) must be operational — you should have contributed to a certification evidence package, not just read the specification. The materials knowledge (task 9) must extend to having made or evaluated a real materials selection decision. TotalCare context (task 11) must be connected to specific life-cycle engineering decisions you have made.

14. Prepare to Walk Through a Complete Engine Sub-System or Component Development — from Requirements Through Certification Evidence 4–6 h

Skill: End-to-end development ownership narrative in a regulated context

Objective: Senior Rolls-Royce engineers are expected to own technical threads that span from requirements definition, through design, through component and rig testing, through engine certification evidence. The interview at this level is a structured probe of whether you have done this — not contributed to it. The thread must include: the specification that governed the design, the key design decisions and their trade-offs, the test plan, the test anomalies encountered, how they were dispositioned, and what the certification authority accepted as evidence of compliance.

Done Signal: For one engine sub-system or major component you personally owned, you can answer questions at every phase without notes — including the hardest disposition decision, the EASA or CAA comment you received on your compliance evidence, and what you would do differently at the design stage to simplify the certification route.

15. Prepare to Discuss a Specific Area Where Your Engineering Judgement Conflicted with a Customer Requirement or a Certification Position — and What You Did 2 h

Skill: Technical authority in a regulated customer-facing context

Objective: Senior Rolls-Royce engineers interface with airline customers on TotalCare-related engineering queries, and with EASA and the CAA on certification matters. The ability to hold a technically defensible position under regulatory or commercial pressure — while maintaining the professional relationship — is a senior differentiator. This is particularly relevant given that Rolls-Royce’s TotalCare model means disputes about engine life limits, maintenance intervals, or unscheduled removals have direct financial consequences for both parties.

Done Signal: You have a story where your engineering position conflicted with what a customer, regulatory authority, or commercial team wanted, you documented and defended your position through the appropriate channels, and the outcome was resolved with a formal engineering record rather than a verbal agreement. The story names the technical substance of the disagreement and the document that captured the resolution. 2 h

16. Build Quantitative Estimation Fluency for Engine Performance Parameters — Without a Calculator for the Setup 4–5 h

Skill: Gas turbine performance reasoning on the spot

Objective: Senior Rolls-Royce engineers are expected to sanity-check analysis results, spot errors in supplier calculations, and respond credibly to “does this number make sense?” in a fast-moving design review. This requires being able to estimate engine performance parameters from first principles: specific thrust, thermal efficiency, pressure ratio implications, specific fuel consumption, turbine entry temperature consequences. Candidates who need a CFD model or performance deck to answer a rough-order-of-magnitude question are demonstrating a gap in engineering intuition that becomes visible at senior level.

Done Signal: You have worked through five gas turbine performance estimation problems (e.g., estimate the thermal efficiency of a given cycle, estimate the compressor work for a given pressure ratio and inlet conditions, estimate the turbine cooling air fraction required for a given turbine entry temperature) — spoken aloud, with assumptions stated before numbers — and can identify which assumption is most load-bearing in each estimate.

17. Prepare a Story of Contributing to or Improving an Engineering Standard, Design Rule, or Corporate Process 2 h

Skill: Institutional knowledge creation and standards authorship

Objective: Rolls-Royce’s engineering depth is encoded in its design rules, material standards, component life limits, and analysis methods — accumulated over decades across the Trent family and earlier programmes. Senior engineers are expected to contribute to this body of knowledge, not just draw on it. Engineers who have spent their career applying standards without ever writing or improving one are demonstrating experience below the level required at senior specialist or technical authority grade.

Done Signal: You have a story where you authored, improved, or formally reviewed an engineering standard, design rule, analytical method, or corporate procedure. The story names the standard or document, the gap it addressed, the evidence base that supported the change, and the approval process it went through before being adopted.

18. Prepare Examples of Mentoring or Developing a Junior Engineer in an Engine-Specific Technical Domain — with a Programme-Level Outcome 1–2 h

Skill: Technical knowledge transfer in a complex, tacit-knowledge-rich environment

Objective: Gas turbine engineering is rich in tacit knowledge — why certain design features exist, what the failure history of a given configuration looks like, which analysis methods are trusted versus which are known to be conservative in specific regimes. This knowledge lives in experienced engineers and is lost when they retire. Senior Rolls-Royce engineers are expected to transfer it actively. The specific question probes whether the junior engineer can now make an independent technical judgement on the relevant topic, not just execute a task.

Done Signal: You have a story where you transferred a specific, engine-domain-relevant technical skill or piece of institutional knowledge to a junior engineer, the junior engineer applied it independently on a real programme, and you can name the programme artefact or design decision that demonstrates the transfer was successful.

1–2 h

All Levels

19. Prepare a Specific Answer to “Describe a Time Safety or Engineering Integrity Came into Conflict with Schedule or Commercial Pressure — What Did You Do?”

1–2 h

Skill: Safety and integrity culture under pressure — the deepest values filter at Rolls-Royce

Objective: Rolls-Royce engines power aircraft carrying hundreds of passengers on every flight. The engineering culture at Derby is explicitly safety-first — not as a value statement but as a commercial necessity: a certification event or an in-service incident on a Trent engine would have consequences that dwarf any schedule saving. Rolls-Royce’s safety and quality councils and engineering gateway processes exist to ensure that safety and engineering integrity decisions are made separately from schedule and commercial pressure. This question appears at every level and the expected answer is one where integrity won at a real cost — not one where both were achieved simultaneously.

Done Signal: You have a story where an engineering integrity position was held against schedule or commercial pressure, the cost was named (delay in days or weeks, cost impact, customer relationship friction), the technical basis for the position was specific, and the outcome confirmed the position was correct or at minimum confirmed the risk was real.

Source: *In Practise*

20. Prepare a Specific, Technically-Grounded View on the Net Zero Challenge for Gas Turbines — Connecting UltraFan, SAF, and Hydrogen to Your Engineering Domain

3 h

Skill: Forward-looking technical positioning on Rolls-Royce’s most consequential strategic challenge

Objective: Rolls-Royce estimates that UltraFan plus 100% SAF could solve roughly 80% of the decarbonisation equation for aviation, making these programmes not just technology demonstrators but the core of the company’s claim on the next generation of widebody and potentially narrowbody platforms. At every level, Rolls-Royce interviewers probe whether candidates have engaged with this challenge technically — not just whether they know it exists. An engineer who can explain what constraints hydrogen combustion places on combustor design, or what SAF composition variability means for fuel system certification, or what the efficiency headroom in the Brayton cycle looks like beyond UltraFan’s target, is demonstrating the kind of technical curiosity Rolls-Royce selects for.

Done Signal: You have a 90-second answer that addresses: what UltraFan’s geared architecture achieves thermodynamically that direct-drive cannot, what 100% SAF changes about combustion chemistry and certification, and one specific engineering challenge in the net zero transition that is directly relevant to your discipline. The answer does not rely on marketing language — every claim in it is technically grounded.

Source: *Aerospace Manufacturing*

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	STAR bank mapped to RR behavioural competencies	All	3–4 h
2	Brayton cycle and engine architecture — derivation level	All	4–5 h
3	10-minute self-presentation as a structured argument	All	3–4 h
4	Physical component identification and reasoning	All	4–5 h
5	Psychometric and SJT test practice	All	4–6 h
6	Live group exercise with observer feedback	All	3–4 h
7	EASA CS-E and Type Certificate framework	Mid+	5–6 h
8	Gate-based development process participation story	Mid+	2 h
9	One advanced engine material in depth	Mid+	4–5 h
10	Reliability or cost improvement story	Mid+	2 h
11	TotalCare model and its engineering consequences	Mid+	2–3 h
12	Supplier non-conformance handling story	Mid+	1–2 h
13	“Why Rolls-Royce?” — technically grounded answer	Mid+	2–3 h
14	Sub-system development from spec through certification	Senior	4–6 h
15	Technical position held under regulatory/commercial pressure	Senior	2 h
16	Gas turbine performance quantitative estimation	Senior	4–5 h
17	Engineering standard or design rule contribution	Senior	2 h
18	Junior engineer mentoring — engine domain, programme outcome	Senior	1–2 h
19	Safety vs. schedule/commercial pressure story	All	1–2 h
20	Net zero technical view — UltraFan, SAF, hydrogen	All	3 h

THE STRUCTURAL REALITY

Rolls-Royce’s business model means it carries engineering accountability for every engine it has ever sold, for as long as it flies. TotalCare does not let Rolls-Royce walk away from a design decision when the engine enters service — it makes every shop visit, every early removal, every life limit extension a direct financial and reputational consequence of the engineering judgements made years earlier. That creates an engineering culture that is simultaneously highly conservative in its standards and deeply attentive to in-service reality.

The candidates who get senior offers are the ones who demonstrate they think about engineering with this whole-life, accountable mindset — not just the design-release mindset that most other engineering organisations cultivate. The test is not whether you know gas turbines. The test is whether you have internalised that every engineering decision you make at Derby will be evaluated not just when the engine passes certification, but when it returns to the shop a decade later — and whether that accountability changes how you approach a trade-off today.